

Luis Eduardo González González

Mexico City | +52 55 6707 6720 | fectda@gmail.com | <https://fectda.github.io/en>

PROFESSIONAL SUMMARY

IT Professional with extensive experience in full-stack development, cloud software architecture, and technology project management. Specialized in Python, JavaScript, and Google Cloud solutions. Recently focused on high-level independent projects involving local generative AI orchestration, advanced observability, and distributed systems. Passionate about building efficient, high-quality solutions, backed by a strong analytical foundation from a background in Physical Sciences.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, Java, Go, Fortran, C, PHP.
- **Web Development (Front-End):** HTML5, CSS3 (SASS), JavaScript, Vue.js, jQuery, Bootstrap, Materialize.
- **Web Development (Back-End):** Python (Flask), PHP (Laravel), Node.js.
- **AI & Automation (New Additions):** Prompt Engineering (Expert Systems), AI Agents (OpenCode, Gemini-CLI), Local LLMs (Ollama, GPU optimization), n8n (AI Automation), Power BI.
- **DevOps & Cloud:** Google Cloud Platform (GCP), AWS (Basics), Docker & Docker Compose (GPU Support), Kubernetes (K8s), Terraform, Ansible, Xen Orchestra.
- **Infrastructure & Networking:** Cloudflare Tunnel, Zero Trust, Kafka, RabbitMQ.
- **Observability:** Grafana, Istio, Prometheus, Traceability, Video Coding.
- **Databases:** SQL (PostgreSQL, MySQL, MariaDB), NoSQL (MongoDB, Redis).
- **Version Control:** Git.
- **Operating Systems:** Linux (Debian), macOS, Windows.
- **Tools & Others:** Arduino, Bash/Shell Scripting, Agile Methodologies (Scrum), Frigate.
- **Data Analysis/Simulation:** Matlab, Mathematica, SPSS, Origin.

PROFESSIONAL EXPERIENCE

Solutions Architect & Maker | Independent Consulting | Mexico City January 2026 – Now

Development of custom solutions in infrastructure, local AI, and physical automation.

- Infrastructure: Orchestration of on-premise servers (Xen Orchestra, K8s, RabbitMQ).
- Applied AI: Implementation of local LLMs (Ollama) for educational and creative use cases.
- Maker: Design and fabrication of custom hardware (3D Printing, IoT).

Research and Development (R&D) | GET IN, S.A.P.I. de C.V. | Mexico City January 2024 – January 2026

- In charge of executing proofs of concept to evaluate the technical and functional viability of new solutions and innovations.
- Completely designed and developed a statistical model for automating data review, resulting in a 90% reduction in manual processing times.
- Continuously formulated hypotheses and developed proofs of concept to validate new technological approaches, actively collaborating with development teams for implementation.

Chief Information Officer (CIO) | GET IN, S.A.P.I. de C.V. | Mexico City June 2018 – January 2024

- Led technology strategy and IT teams, covering product feature development, inventory management, support, and process design.
- Directed the infrastructure migration to Google Cloud Platform (GCP) implementing PaaS solutions, significantly reducing maintenance costs and increasing service reliability.
- Driven the re-engineering of system architecture towards a microservices model, facilitating evolutionary maintenance and scalability.
- Redesigned data processing systems into a horizontal structure, cutting processing times to one-third and reducing associated operating costs by one-third.
- Managed 5 multidisciplinary teams: Front-End, Back-End, Embedded Software, Data Engineering, and Warehouse.
- Responsible for requirements definition, staff technical training, and critical incident resolution.
- Key Technologies: Go, Python, PHP (Laravel), MongoDB, Java, Vue.js, GCP.

Senior Developer | elcultivo.mx | Mexico City February 2015 – June 2018

- Transformed the company's focus from portfolio web design to custom CRM application development for e-commerce.
- Established and led new development processes, successfully delivering 20 distinct web applications for 20 different clients in one year.
- Responsible for software architecture design (ER Models, MVC with Laravel and Vue.js).
- Led the development team, managing requirements gathering and technical training.
- Collaborated closely with graphic designers on UI/UX optimization.
- Key Technologies: PHP (Laravel), Vue.js, JavaScript, MySQL, MongoDB, HTML5, CSS3, Redis.

PAIMEF Monitoring Liaison | National Institute for Social Development (Indesol) | Mexico City 2013 – 2015

- Responsible for generating technical mechanisms for evaluating PAIMEF resources.
- Identified inefficiencies in KPI tracking and proposed a centralized web system.
- Proactively designed and developed two interconnected web systems (PHP/MySQL) to automate data collection and analysis.
- Implemented and maintained successive versions, significantly improving monitoring efficiency.

INDEPENDENT PROJECTS & FREELANCE (Current)

Infrastructure, Cloud & Observability (2026 - Present):

- **Local Server Orchestration:** Management of on-premise infrastructure using Xen Orchestra, Terraform, and Ansible for automated provisioning.
- **Scalable Distributed Systems:** Development of a Kubernetes (K8s) message processing system integrated with RabbitMQ. Implementation of pod autoscaling based on workload and pipelines where each worker runs independent jobs.
- **Security & Networking:** Implementation of Cloudflare Tunnel and Zero Trust policies for secure resource access.
- **Advanced Observability:** Deployment of monitoring stacks with Grafana, Istio, and Prometheus; implementation of distributed tracing for microservices.

Generative AI & Automation:

- **Language Learning Platform (AI Roleplay):** Development of a local dockerized application utilizing LLMs (via Ollama with GPU optimization) to simulate a language instructor. The system manages complex role-play interactions via voice and text.

- **AI Agents & Workflows:** Creation and programming of autonomous agents using OpenCode and Gemini-CLI. Automation of complex business processes using intelligent workflows in n8n.
- **Advanced Prompt Engineering:** Design of "Expert Systems" using advanced prompting techniques for specialized tasks.

Simulations & Home Automation (Historical):

- **Physical Simulations:** Developed interactive simulations for planetary motion (Basic) and gravitational waves (Fortran).
- **Neural Networks:** Experimented with development and training of neural networks for image classification using Python/TensorFlow.
- **Advanced Home Automation:** Implemented control systems (lighting, temperature, security) with Home Assistant and Frigate.
- **3D Design:** Architectural and technical part modeling for 3D printing (SketchUp, FDM).

EDUCATION

- **M.S. in Physics** | National Autonomous University of Mexico (UNAM) | Mexico City
 - Specialization in Mesoscopic Nonequilibrium Thermodynamics | 2008 – 2010
- **B.S. in Physics** | **National Autonomous University of Mexico (UNAM)** | **Mexico City** | **2003 – 2007**
- **Diploma in Results-Based Budgeting (RBB)** | UNAM / SHCP (Ministry of Finance) | 2014

TEACHING EXPERIENCE

Guest Lecturer, Master in Web Development | Universidad Andina Simón Bolívar | 2018.

- Specialized module on ORM usage (Eloquent/Laravel).
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LANGUAGES

- Spanish: Native
- English: B2 Level (Professional Working Proficiency)

PUBLICATIONS / MISCELLANEOUS

- **Co-author:** "Experiments with Soap Bubbles" (FENOMECE Notes, science outreach publication).
- Electrical Technician skills.
- PC and electronics repair.
- Advanced proficiency in office suites (Microsoft Office Suite, Google Workspace).